Network Intrusion Detection System (IDS) Using Snort

CodeAlpha Cyber Security Internship – Task 4

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Abstract

This project demonstrates the implementation of a Network Intrusion Detection System using Snort on Kali Linux. A custom ICMP detection rule was created and tested successfully.

Objectives

Install Snort, configure custom rules, monitor traffic, generate ICMP packets, and verify alert generation.

Tools Used

Kali Linux, Snort 3.12.2.0, Nano Editor, Terminal, Ping Utility.

Implementation

The following screenshots document the complete implementation process.

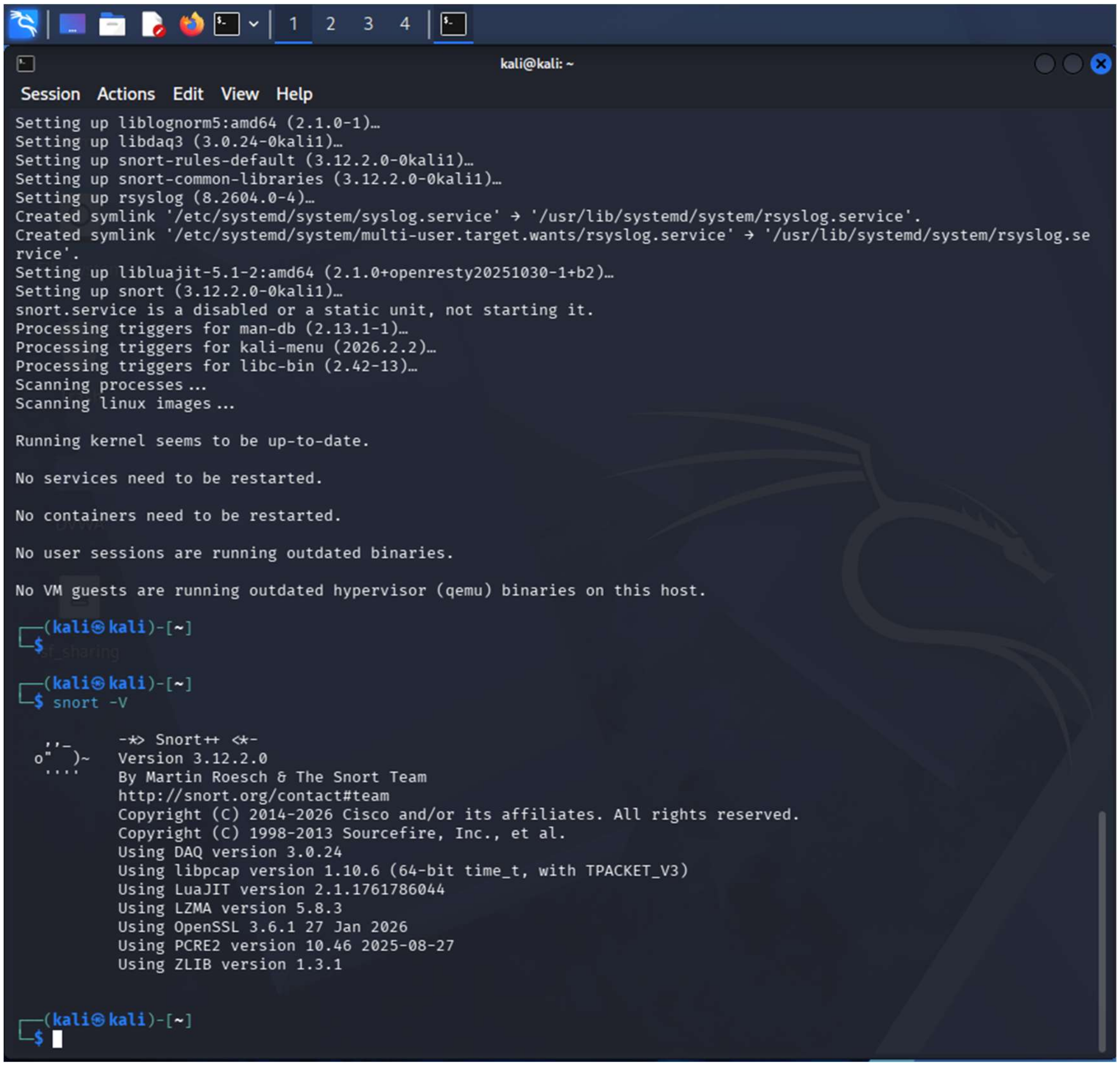

Figure 1: Snort Installation and Version Verification

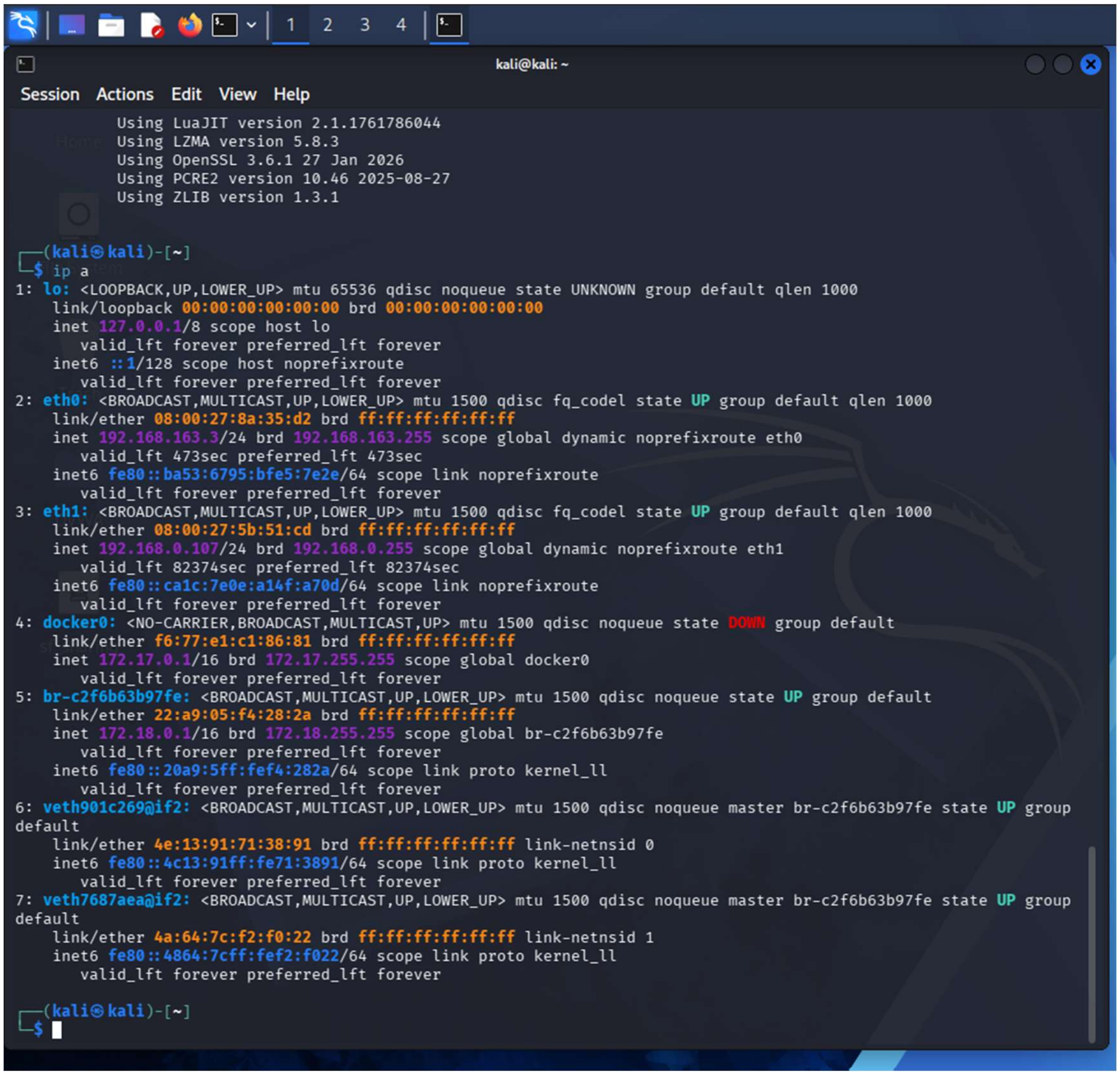

Figure 2: Network Interface Identification (eth0)

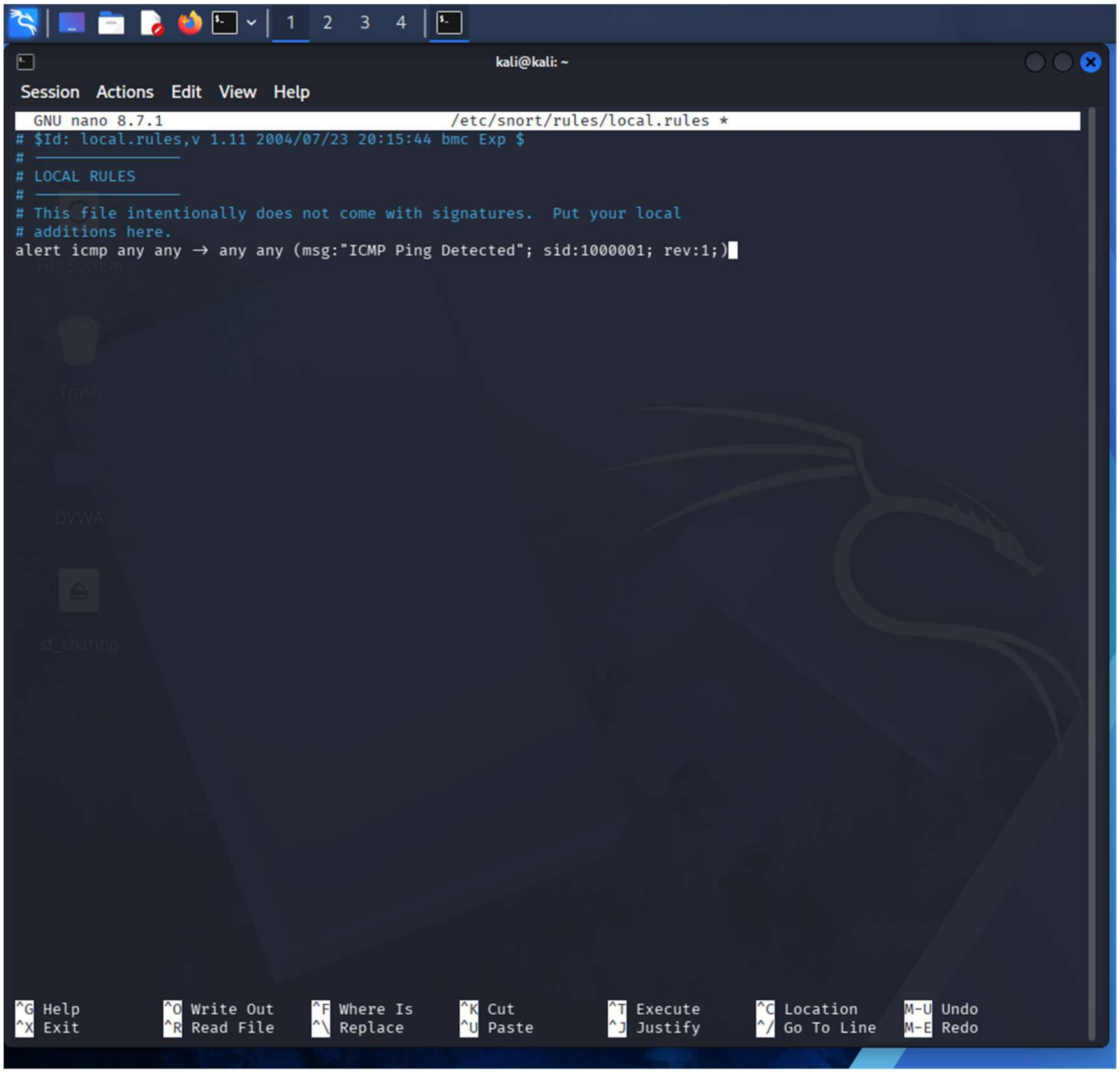

Figure 3: Custom ICMP Detection Rule

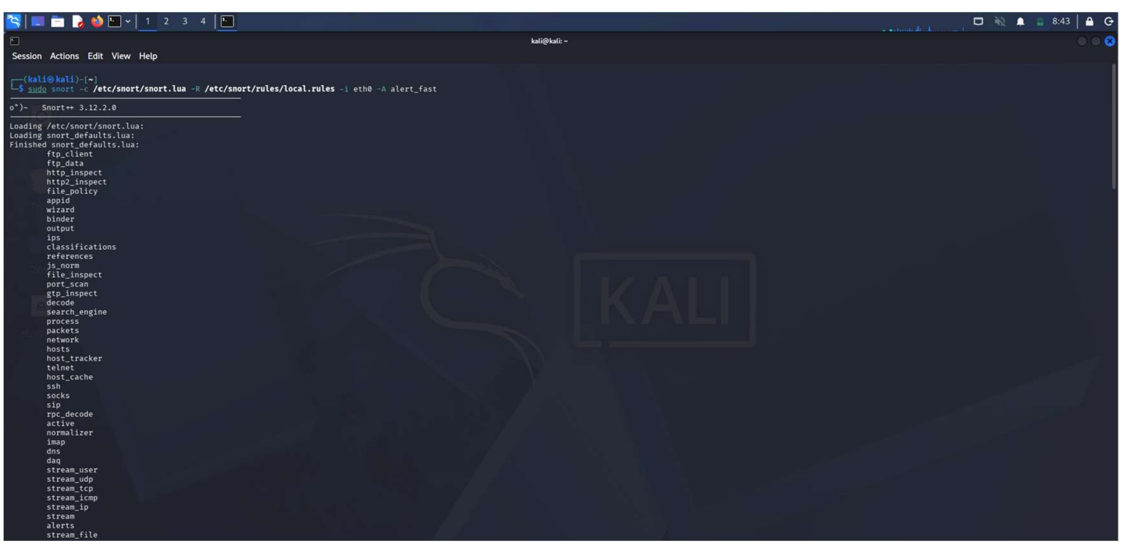

Figure 4: Snort Monitoring Started

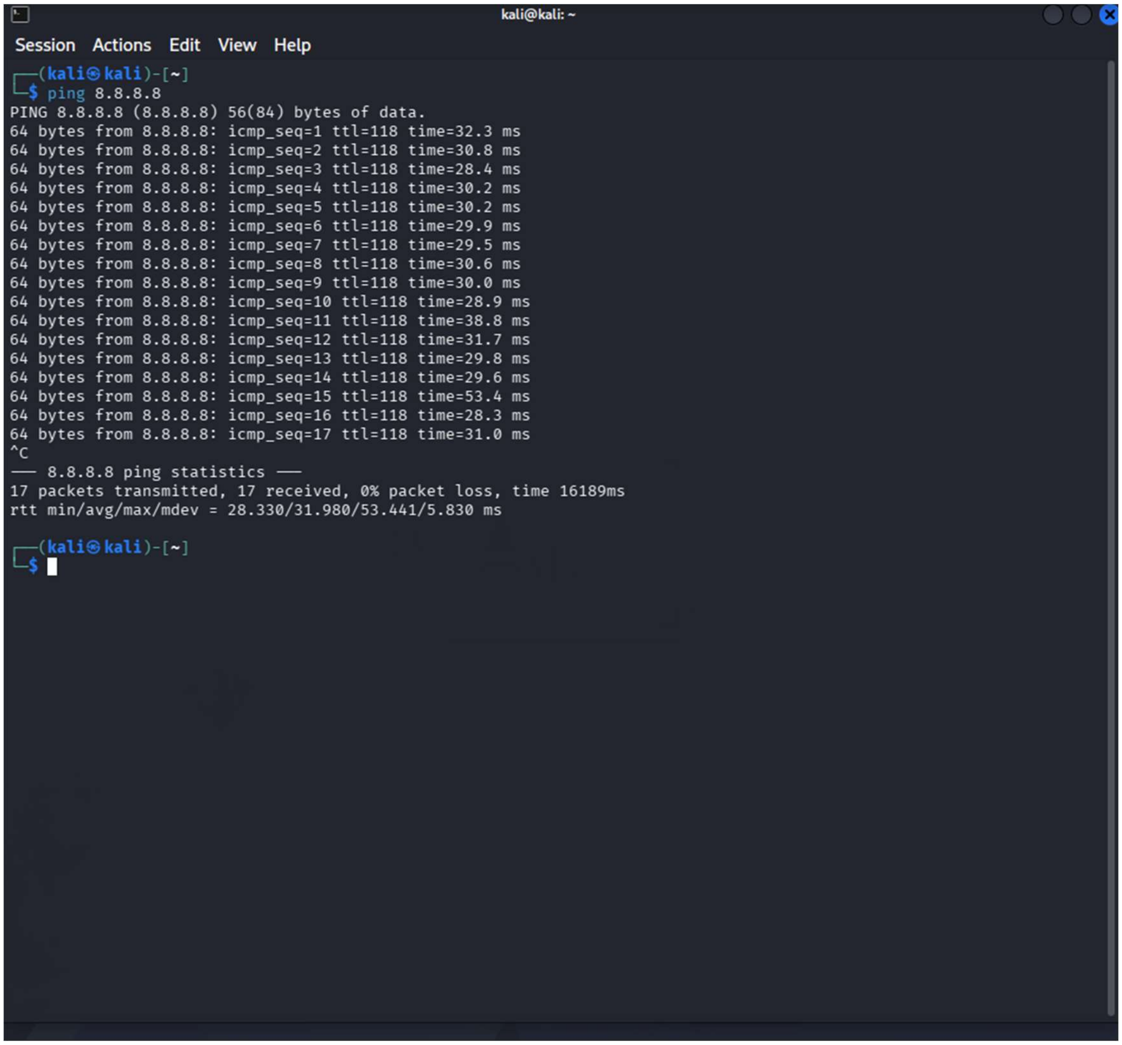

Figure 5: ICMP Traffic Generation using Ping

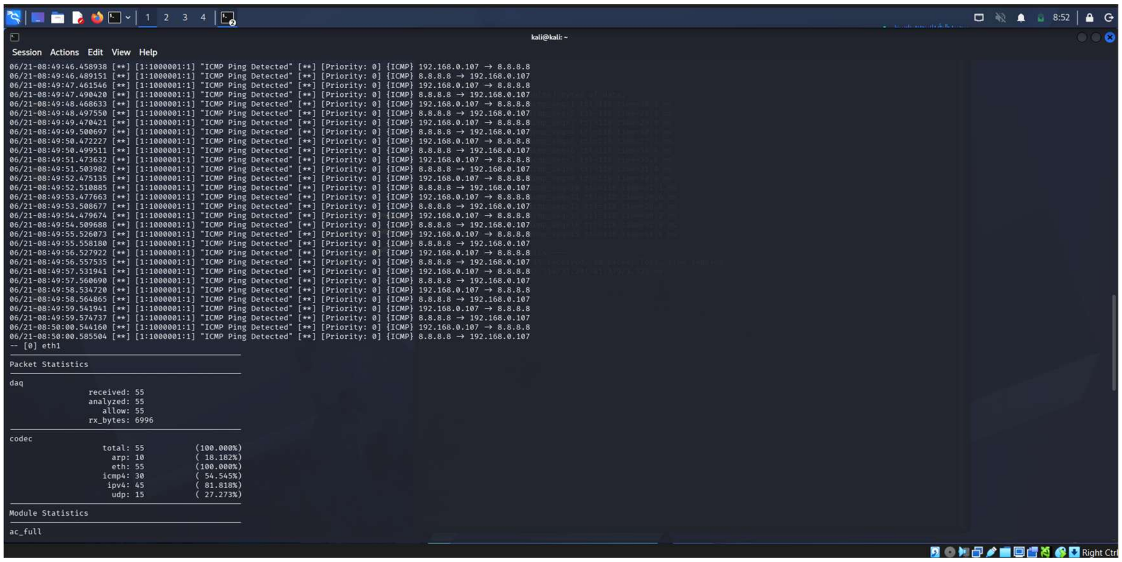

Figure 6: Snort Alert Detection Output

Results

Snort successfully detected ICMP traffic generated using the ping command. The custom rule triggered alerts confirming correct IDS operation.

Conclusion

The Network IDS was successfully implemented and tested. Snort detected ICMP packets and generated alerts according to the configured rule.